MA388 Sabermetrics: Lesson 29

Home Run Hitting II

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Review

What characteristics of a hitter does OPS quantify?

Why might one use OPS+ or adjusted OPS+ instead of OPS?

What are some limitations of these statistics?

Linear Weights

Recall we talked about the relationship between Runs and Wins in Chapter 4 of the Marchi text. What was the rule of thumb we used?

(*Understanding Sabermetrics* page 83) Ideally, we want a statistic for batters that:

- quantifies their contribution to runs scored.
- is not based upon the situations the batters faced when they came to the plate (since their batting actions did not create those situations).

How about RBIs? A batter is awarded Runs Batted In (RBIs) for most situations when his plate appearance results in runs scored. Why are RBIs not a good measure of a player's contribution to runs scored?

Various researchers have proposed linear models to quantify a player's contribution to runs scored. These models give weights to various individual statistics. For the purposes of today's lesson, let's investigate the condensed model proposed by Thorn and Palmer (*Understanding Sabermetrics*, page 87). I'll refer to this model as the Linear Weights Model:

$$BattingRuns = w_1 1B + w_2 2B + w_3 3B + w_4 HR + w_5(BB + HBP) - w_6(AB - H)$$

How can we get weights for the Linear Weights Model?

$$RV = EXP_{New} - EXP_{Old} + RunsScored$$

How do we interpret BattingRuns?

Let's see what this looks like for the 2011 season.

```
library(Lahman)
library(tidyverse)
library(plotly)
library(knitr)

# Calculate weights from run values using Retrosheet play-by-play data.
# We're getting this file from GitHub in the linear_weights() function,
# but you can adapt it to a Retrosheet file you have locally if desired.

source("linear_weights.R")
weights <- linear_weights(2011) |> pluck("weights")
weights |> kable(digits = 3)
```

Event	weight
1B	0.442
2B	0.736
3B	1.064
BB.HBP	0.289
HR	1.392
Out	-0.255

How do these weights compare to those in slugging percentage (SLG)?

```r
# Filter for 2011 Players with at least 500 at-bats.
vars = c("AB", "H", "X2B", "X3B", "HR", "BB", "HBP", "SF", "RBI")
Batting |>
  filter(yearID == 2011) |>
  group_by(playerID) |>
  summarise_at(vars, sum) |>
  filter(AB >= 500) |>
  left_join(People |> select(nameLast, nameFirst, playerID) )|>
  mutate(name = paste(nameFirst, nameLast, sep = " "))|>
  select(name, everything(), -nameLast, -nameFirst) |>
  mutate(X1B = H - X2B - X3B -HR,
         SLG = (X1B + 2*X2B + 3*X3B + 4*HR) / AB,
         OBP = (H + HBP + BB) / (AB + HBP + SF + BB),
         OPS = SLG + OBP,
         AVG = H / AB) -> batting_2011

# Calculates Batting Runs using Thorn and Palmer's condensed Linear Weights model.
# Note: Statistics and weights have to be in the same order.
batting_runs_fn <- function(statistics, weights){
  runs <- round(sum(statistics * weights), 1)
  return(runs)
}

batting_2011 |>
  group_by(playerID) |>
  mutate(batting_runs = batting_runs_fn(statistics = c(X1B, X2B, X3B,
                                                       BB + HBP, HR, AB - H),
                                        weights = weights |> pull(weight))) |>
  arrange(-batting_runs) -> batting_2011


batting_2011 |>
  select(name, AB, H, HR, RBI, AVG, OPS, batting_runs) |>
  head(10) |>
  kable(digits = 3,
        caption = "Top 10 MLB Players (with at least 500 at bats) - Batting Runs 2011")
```

Table 2: Top 10 MLB Players (with at least 500 at bats) - Batting Runs 2011

playerID	name	AB	H	HR	RBI	AVG	OPS	batting_runs
bautijo02	Jose Bautista	513	155	43	103	0.302	1.056	66.4

playerID	name	AB	H	HR	RBI	AVG	OPS	batting_runs
cabremi01	Miguel Cabrera	572	197	30	105	0.344	1.033	66.3
kempma01	Matt Kemp	602	195	39	126	0.324	0.986	54.9
fieldpr01	Prince Fielder	569	170	38	120	0.299	0.981	54.7
gonzaad01	Adrian Gonzalez	630	213	27	117	0.338	0.957	51.9
braunry02	Ryan Braun	563	187	33	111	0.332	0.994	51.4
vottojo01	Joey Votto	599	185	29	103	0.309	0.947	50.5
ellsbja01	Jacoby Ellsbury	660	212	32	105	0.321	0.928	44.3
ortizda01	David Ortiz	525	162	29	96	0.309	0.953	41.9
grandcu01	Curtis Granderson	583	153	41	119	0.262	0.916	39.0

How many wins would you attribute to Matt Kemp in the 2011 season?

```r
batting_2011 |>
  select(name, AB, H, HR, RBI, AVG, OPS, batting_runs) |>
  tail(10) |>
  kable(digits = 3,
        caption = "Bottom 10 MLB Players (with at least 500 at bats) - Batting Runs 2011")
```

Table 3: Bottom 10 MLB Players (with at least 500 at bats) - Batting Runs 2011

playerID	name	AB	H	HR	RBI	AVG	OPS	batting_runs
pierrju01	Juan Pierre	639	178	2	50	0.279	0.657	-14.8
wellsve01	Vernon Wells	505	110	25	66	0.218	0.660	-15.2
desmoia01	Ian Desmond	584	148	8	49	0.253	0.656	-15.7

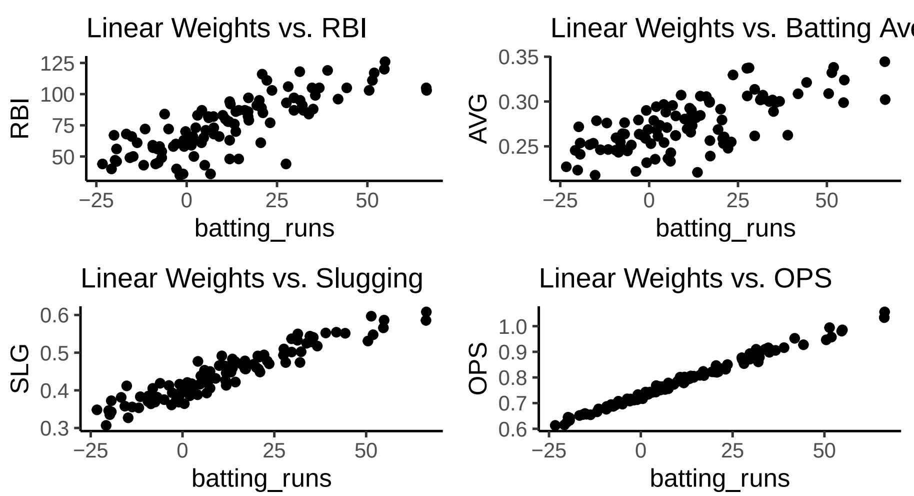
playerID	name	AB	H	HR	RBI	AVG	OPS	batting_runs
betanyu01	Yuniesky Betancourt	556	140	13	68	0.252	0.652	-16.7
escobal02	Alcides Escobar	548	139	4	46	0.254	0.633	-19.4
gonzaal02	Alex Gonzalez	564	136	15	56	0.241	0.642	-19.4
suzukic01	Ichiro Suzuki	677	184	5	47	0.272	0.645	-19.8
mcgehca01	Casey McGehee	546	122	13	67	0.223	0.626	-20.1
bartlja01	Jason Bartlett	554	136	2	40	0.245	0.615	-20.8
riosal01	Alex Rios	537	122	13	44	0.227	0.613	-23.3

Let's see how BattingRuns compares to traditional statistics.

```
library(gridExtra)
title_size = 12
plot_rbi <- batting_2011 |>
  ggplot(aes(label = name,
             x = batting_runs,
             y = RBI)) +
  geom_point() +
  labs(title = "Linear Weights vs. RBI") +
  theme_classic() +
  theme(plot.title = element_text(size = title_size))

plot_avg <- batting_2011 |>
  ggplot(aes(label = name,
             x = batting_runs,
             y = AVG)) +
  geom_point() +
  labs(title = "Linear Weights vs. Batting Average") +
  theme_classic() +
  theme(plot.title = element_text(size = title_size))
```

```
plot_slg <- batting_2011 |>
  ggplot(aes(label = name,
             x = batting_runs,
             y = SLG)) +
  geom_point() +
  labs(title = "Linear Weights vs. Slugging") +
  theme_classic() +
```

```
  theme(plot.title = element_text(size = title_size))

plot_ops <- batting_2011 |>
  ggplot(aes(label = name,
             x = batting_runs,
             y = OPS)) +
  geom_point() +
  labs(title = "Linear Weights vs. OPS") + theme_classic() +
  theme(plot.title = element_text(size = title_size))

grid.arrange(plot_rbi, plot_avg, plot_slg, plot_ops, ncol = 2)
```



Let's say you're a general manager. Which statistic would you use?

What other factors would you want to consider?

In terms of BattingRuns, is it more important to hit for power or on base percentage?

```
batting_2011 |>
  ungroup() |> # Before adding this, scale() returned NaN for everything.
  mutate(OBP_scaled = scale(OBP),
         SLG_scaled = scale(SLG),
         BR_scaled = scale(batting_runs)) |>
  ggplot(aes(x = OBP_scaled, y = SLG_scaled, color = BR_scaled)) +
  geom_point() +
  scale_color_viridis_b() +
  geom_abline(slope = 1, intercept = 0)
```